

DecisionIQ Master Handbook

CHAPTER 7: The Technology Cast: Roles of Python, SQL, and Machine Learning

7.1 Tech Stack Mapping

The technologies in DecisionIQ are selected for performance, maintainability, and clean separation of concerns.

DecisionIQ Tech Stack Role
Streamlit & Plotly (UI & Interactive Visuals)
Scikit-Learn (Churn & Supplier Risk ML Classifiers)
SQL & SQLAlchemy (Database Logic & Object Relational Map)
Pandas & Polars (Data Cleaning & Feature Engineering)
Python (Core System Orchestration)

- **Python:** The orchestrator used to run the data generation scripts, pipeline validations, and model training tasks.
- **Pandas & Polars:** Used for data manipulation, normalization, and memory-efficient feature engineering.
- **SQLAlchemy:** The Object-Relational Mapper (ORM) used to abstract SQL connections, allowing database-independent queries.
- **SQLite & PostgreSQL:** Storage engines. SQLite serves as the default zero-config local engine, and PostgreSQL acts as the production target.
- **Advanced SQL (ANSI Compliant):** Used to write high-performance cohort and analytical queries directly inside the database.
- **Scikit-Learn:** Used to train predictive Random Forest and Ridge Regression models.
- **Plotly:** Used to render interactive charts (waterfall, scatter, line charts) with hover states.
- **Streamlit:** Used to render the final browser-based control panel.